

Midterm Exam Review

CSE3031 Computer Architecture

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What's Covered?

- Fundamentals of Computer Design
- Instruction set principles and examples
- Memory hierarchy

Fundamentals of Computer Design

- Performance Metrics
- Processor Performance Equation
 - IC, CPI, cycle time
- Amdahl's Law

Instruction Set Principles and Examples

- Instruction set architectures
 - CISC vs RISC
- Seven dimensions of ISA
 - Class of ISA
 - Memory addressing
 - Addressing modes
 - Types and sizes of operands
 - Operations
 - Control flow instructions
 - Encoding an ISA
- ISA Example: MIPS64

Memory Hierarchy

- Cache fundamentals
 - Block placement
 - Block identification
 - Writing strategies
 - Block replacement
- Cache performance
 - Average memory access time: hit time, miss rate, miss penalty
 - Improve cache performance
 - Miss penalty: multi-level of caches (local and global miss rates)
 - Miss rate: 3C's

Sample Question 1

- Assume for a load-store ISA, the percentage of load instructions is 25% and the percentage of store instructions is 10%. Find
 - The percentage of all memory accesses for instruction
 - The percentage of all memory accesses that are reads

Solution

- The percentage of all memory accesses for instruction: $1/(1+0.35) = 74.1\%$
- The percentage of all memory accesses that are reads: $(1+0.25)/(1+0.35) = 92.6\%$

Sample Question 2 - HW1

Q3

- Consider two implementations M1 and M2 of the same ISA. We are interested in the performances of two programs P1 and P2, which have the following instruction mixes:

<i>Operations</i>	<i>P1</i>	<i>P2</i>
Load/store	40%	50%
ALU	50%	20%
Branches	10%	30%

- The CPIs for each machine are:

<i>Operations</i>	<i>M1</i>	<i>M2</i>
Load-store	2	2
ALU	1	2
Branches	3	2

- (a) Assume that the clock rate of M1 is 2 GHz. What should be the clock rate of M2 so that both machines have the same execution time for P1?
- (b) Assume now that both machines have the same clock rate and that P1 and P2 execute the same number of instructions. Which machine is faster for a workload consisting of equal runs of P1 and P2?
- (c) Find a workload (using only P1 and P2) that makes M1 and M2 have the same performance when they have the same clock rate.

Answer (a)

- Using P1's instruction mix:
- $CPI_1 = 40\% * 2 + 50\% * 1 + 10\% * 3 = 1.6$
- $CPI_2 = 40\% * 2 + 50\% * 2 + 10\% * 2 = 2$
- $ClockFreq_1 = 2GHz$
- Instruction counts are the same for M1 and M2
- $IC_1 * CPI_1 / ClockFreq_1 = IC_2 * CPI_2 / ClockFreq_2$
- $ClockFreq_2 = 2 * 2 / 1.6 = 2.5GHz$

Answer (b)

- Considering 1 run of P1 and 1 run of P2, the instruction count for P1 and P2 is I, cycle time for both M1 and M2 is T.
- For M1,
 - $\text{CPI_P1} = 40\% \cdot 2 + 50\% \cdot 1 + 10\% \cdot 3 = 1.6$
 - $\text{CPI_P2} = 50\% \cdot 2 + 20\% \cdot 1 + 30\% \cdot 3 = 2.1$
 - Execution time of M1 = $I \cdot \text{CPI_P1} \cdot T + I \cdot \text{CPI_P2} \cdot T = 3.7 \cdot I \cdot T$
- For M2,
 - $\text{CPI_P1} = 40\% \cdot 2 + 50\% \cdot 2 + 10\% \cdot 2 = 2$
 - $\text{CPI_P2} = 50\% \cdot 2 + 20\% \cdot 2 + 30\% \cdot 2 = 2$
 - Execution time of M2 = $I \cdot \text{CPI_P1} \cdot T + I \cdot \text{CPI_P2} \cdot T = 4 \cdot I \cdot T$
- M1 is faster.

Answer (c)

- Considering the workload contains 1 run of P1 and n runs of P2
- For M1,
 - Execution time of M1 = $I * CPI_{P1} * T + n * I * CPI_{P2} * T = (1.6 + 2.1 * n) * I * T$
- For M2,
 - Execution time of M2 = $I * CPI_{P1} * T + n * I * CPI_{P2} * T = (2 + 2 * n) * I * T$
- Execution time of M1 = Execution time of M2
 - $(1.6 + 2.1 * n) * I * T = (2 + 2 * n) * I * T$
 - $n = 4$
- The workload contains 1 run of P1 and 4 runs of P2

Sample Question 3 - HW1

Q4

- Assume the sequential execution occurs in 15% of the program, i.e. other 85% of the program can be parallelized with N processor with a speedup of N :
- (a) What is the maximum overall speedup with an infinite number of processors?
- (b) How many processors are required to be within 20% of the maximum speedup?
- (c) How many processors are required to be within 2% of the maximum speedup?

Answer

- (a) Using Amdahl's law
- $\text{Speedup} = 1/(1 - f + f/s)$
- $f = 0.85$, $s = \text{infinity}$ (for an infinite number of processors)
- $\text{Speedup} = 1/(0.15 + 0.85/\infty) = 6.67$

- (b) Assume N processors are required to be within 20% of maximum speedup
- $80\% * 6.67 = 1/(0.15 + 0.85/N)$
- $N = 23$

- (c) Assume N processors are required to be within 2% of maximum speedup
- $98\% * 6.67 = 1/(0.15 + 0.85/N)$
- $N = 278$

Sample Question 4 - HW1

Q5

- Availability is the most important consideration for designing servers, followed closely by scalability and throughput.
- (a) We have a single processor with a failure in time (FIT) of 100. What is the mean time to failure (MTTF) for this system?
- (b) If it takes one day to get the system running again, what is the availability of the system?
- (c) Imagine that the government, to cut costs, is going to build a supercomputer out of inexpensive computers rather than expensive, reliable computers. What is the MTTF for a system with 1000 processors mentioned in (a)? Assume the processor has the exponentially distributed lifetime and if one processor fails, the system fails.

Answer

- (a) $MTTF = 1/FIT = 10^9 \text{ hours}/100 = 10^7 \text{ hrs}$
- (b) $MTTR = 1 \text{ day} = 24 \text{ hrs}$, $\text{Availability} = MTTF/(MTTF + MTTR) = 10^7/(10^7 + 24) = 99.99976\%$
- (c) $FIT_{sc} = 1000 * FIT = 10^5$
- $MTTF_{sc} = 1/FIT_{sc} = 10^9 \text{ hours}/10^5 = 10^4 \text{ hrs}$

Sample Question 5

Consider a processor with an instruction length of 10 bits and with 16 general-purpose registers.

(1) How many bits are needed to encode a register field?

(2) Is it possible to have instruction coding for the following:

- 3 two-address instructions, 14 one-address instructions, and 31 zero-address instructions
- 3 two-address instructions, 15 one-address instructions, and 17 zero-address instructions
- Please justify your answer.

Answer

1. 4 bits

2.

(a) Yes

(b) No, max 3 two, 15 one, 16 zero

Sample Question 6

- Consider a system in which we have two 64KB split caches, and a system in which we have one 128KB unified cache
- 40% of instructions are Loads or Stores
- A hit takes 1 clock cycle
- A Load or Store takes 1 extra clock cycle when dealing with the single cache (which is a structural hazard)
- Assumed that the measured misses per instructions are:
 - 64KB Instruction: 8 per 1000 instructions
 - 64KB Data: 40 per 1000 instructions
 - 128KB Unified: 45 per 1000 instructions
- Question: What is the value of the miss penalty, P , in number of cycles, for which both systems are equivalent in terms of performance?
 - Hint: First compute the miss rates for all

Sample Question 6

- Miss rates:
 - 64KB Instruction
 - 8 misses per 1000 instructions
 - Each instructions corresponds to 1 instruction memory access
 - miss rate = $8/1000 = 0.008$
 - 64KB Data
 - 40 misses per 1000 instructions
 - 40% of instructions correspond to 1 data memory access
 - miss rate = $40 / (1000 * .40) = 0.1$

Sample Question 6

- Miss rates:
 - 128KB Unified
 - 45 misses per 1000 instructions
 - 60% of instructions correspond to 1 instruction memory access
 - 40% if instructions corresponds to 1 instruction and 1 data memory access
 - miss rate = $45 / (1000 * 1 * 0.60 + 1000 * 2 * 0.40) = 0.0321$
- Now we must compute memory access time

Sample Question 6

- $\text{MAT} = \text{hit time} + \text{miss rate} \times P$
- We need to know the fraction of memory accesses that are for instructions and the fraction for data
 - For X instructions: X I accesses, $.4X$ D accesses
 - Fraction of I accesses: $X / 1.4X = 71\%$
 - Fraction of D accesses: $.4X / 1.4X = 29\%$

Sample Question 6

- Split caches (I: 0.008, D: 0.1)
 - $\text{MAT} = .71 (1 + 0.008P) + .29 (1 + 0.1P)$
 $= 1 + 0.0346P$
- Unified cache (0.0321)
 - $\text{MAT} = .71 (1 + 0.0321P) + .29 (1 + 1 + 0.0321P)$
 $= 1.29 + 0.0321P$
- The two are equiv. if: $1 + 0.0346P = 1.29 + 0.0321P$
 $\Rightarrow P = 116 \text{ cycles}$